Supplement: methods, sensitivities and correction record

Second draft for review, 10 November 2026. No release or new data access is authorised. Barry W. Brook*[1]

Draft for review; release and submission are Barry W. Brook's decision.

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This supplement distinguishes the frozen original comparison, the exploratory observation and clock reassessment, synthetic precision planning, and the prospectively frozen transfer screen. Every quoted number is linked to the claims ledger, `manuscript/claims.csv`. The interpretation specification, `manuscript/estimands.md`, states what each analysis bounds and which phrasings are not allowed. Section and decision labels (A04, A10, D018 and so on) refer to the project record in `docs/` and `coordination/decisions.md`.

S1. Definitions, readings and units

Term	Meaning
Issuance	One forecast, issued when the observation-time rule first confirms that an outburst has ended
Duration	Onset-to-end time of the completed outburst, from X4 run on the 400 days before issue
Quiet gap	End of one outburst to onset of the next; fitted-end and observed-anchor proxies are distinguished in S2.8
Previous-gap predictor, q	Standardised log of the preceding quiet gap, used only if covered (no visit gap over seven days)
Recent-level predictor, L	Standardised mean of up to six earlier covered log quiet gaps; zero after standardisation if none
Fitted onset interval	X4's retrospective crossing summary; working scoring evidence, not a known physical bracket
Right censoring	The forecast ends at coverage loss, the next issuance, or the block end before an onset is established
Working interval score	Log forecast mass for the fitted onset interval, or log survival to a selected cutoff, conditioned on issue; a proper latent-onset likelihood is unestablished
Paired gain	M3's log score minus M2's for the same issuance
Geometric mean likelihood ratio	exp(mean paired gain); a 1.7% increase for the primary
Working clock scale	The median of the untruncated lognormal clock; ratios of scales are model quantities, not observed mean waits
Partial-identification bounds	The smallest and largest coverage consistent with bracketed or censored outcomes

Times are Julian Dates as returned by the archive, without conversion. Durations and gaps are in days. Log durations are standardised by the frozen development centre and scale.

S2. Methods

S2.1 Observations and extraction

Rows were requested in bounded annual windows, ending strictly before 2018. Responses near the 50,000-row service cap were bisected, and every response was hashed on receipt before parsing. The analysis uses rows in the `Vis.` band with validation flag `V`. The 1960–2017 evaluation window contains 63 validated leaf responses. These hold 500,479 rows after removing 81 duplicate identifiers, of which 249,356 meet the analysis selector.

X4 (`src/dnmem/extract_x4.py`, definition `a04f-x4-v1`) groups observer visits into candidate episodes brighter than visual magnitude 10. For each episode it forms normalised slope-mixture summaries for the brightening and fading threshold crossings, using nearby estimates plus a shared 0.05-mag systematic level-error allowance. Plug-in noise, slope weighting and clipping do not establish a calibrated posterior or known containment of physical onset. The allowance prevents unlimited shrinkage as estimates accumulate; it was chosen on development simulations, not measured as the true shared observer error. An episode is confirmed by bright reports from two distinct observers, or from one observer on at least two nights with a margin past the threshold and no contradicting faint report. The observation-time confirmation rule arms only on bright evidence at least 0.5 mag past the threshold. Reports closer to the threshold cannot arm it; noisy reports that cross the margin can still do so.

Three earlier extractors, X1 to X3, failed qualification. X4 passed a synthetic screen at magnitude 10 on the qualified phase-aligned cadence family, and the C1/7 d coverage rule was selected there. The unseen waveform family failed benchmark qualification before X4 recovery was scored; X4 recovery on that family was not tested. Magnitude-9.5 diagnostics failed in later eras. Those failures are retained (A04e–A04i). The evaluation is therefore conditional on the magnitude-10 X4 extraction.

S2.2 Issuance and predictors

Each observation-time end confirmation issues one forecast at its emission time. A two-day reporting delay is a sensitivity. Every emission is kept, including false and repeat emissions. Predictors come from X4 run on the rows in the 400 days up to the emission and on nothing else. A unit test inserts poisoned rows after the emission and checks that no ledger field changes. The current episode is the last prefix episode whose end bracket reaches the emission's bright evidence. A duration of zero or less makes the mark unavailable; nothing is clipped. The previous gap requires the previous episode to begin more than 30 days inside the window.

S2.3 Outcomes and censoring

The original outcome is the next onset summary found by retrospective X4 up to an explicit cutoff. An onset bracket that straddles issue does not establish survival, and that forecast is excluded under the frozen rule. Here none were excluded. A forecast is scored on the onset interval if the interval is established before the first of these events:

- coverage loss, the first visit gap longer than seven days after issue;
- supersession by the next issuance;
- the end of the block.

Otherwise the forecast is right-censored at the earliest of these events.

S2.4 Models and working scores

Each model is a lognormal clock for the quiet gap Q after a fitted end anchor e, with log-location μ and log-standard deviation σ:

- M0: $\mu = \alpha$;
- M2: $\mu = \alpha + \gamma_1 q + \gamma_2 L$;
- M3: $\mu = \alpha + \gamma_1 q + \gamma_2 L + \beta x$, with $\beta \geq 0$.

Here $q = (\log Q_{\text{prev}} - c_q)/s_q$. If ℓ is the mean of up to six earlier covered log quiet gaps, $L = (\ell - c_L)/s_L$; set $L = 0$ if none are available. The centres and scales are frozen development means and standard deviations. Likewise, x is the standardised log duration. At issue age a, an onset bracket $(l, u]$ scores $\log P(l - e < Q \leq u - e) - \log S(a)$. A censoring time c scores $\log S(c - e) - \log S(a)$, where S is the survival function. Forecasts without a covered previous gap use M0 in both models, and forecasts without an available fitted end use a common remaining-wait forecast. Both contribute exactly zero to the contrast.

M2 was fitted on timing-eligible 1910–1959 rows and frozen. The duration offset was then fitted on rows with a valid mark, with M2 held fixed, so that $\beta = 0$ reproduces M2 exactly (D021). Frozen values (C04): $\beta =$ 0.046597 per standard deviation of log duration; previous-gap slope 0.078016; recent-level slope 0.051336; σ = 0.34636. The parameter file's SHA-256 begins `ce4ef8d5`.

The constraint $\beta \geq 0$ and matched routing respond to a measurement coupling. The duration and the following gap share the estimated end time, so end-time error couples them. In earlier synthetic work that coupling pushed fitted null coefficients negative (D020). Codex then showed that a nonnegative bound is a restriction of direction, not a theorem about the sign of the error. An exact construction with constant true duration can yield positive information (D021). Endpoint coupling therefore remains a stated limit. The uniform-end and two-day-delay sensitivities probe it only partly.

S2.5 Primary statistic and inference

The statistic is the mean paired gain over all score-eligible issuances with issue times in JD 2436934.5 to 2458119.5. The standard error is cluster-robust, over consecutive clusters of 6×365.25 days anchored at the window start, with the factor $G/(G-1)$ for G nonempty clusters. Inference uses Student's t with $G - 1$ degrees of freedom. The result counts as confirmed if the one-sided 95% lower bound exceeds zero. The cluster length was chosen before any operating characteristics were seen (D021). There is one confirmatory statistic. No subset, era, variant or interval width may replace it.

S2.6 Operating characteristics

Before the development fit, a finite synthetic screen checked the design (A10d). At an effect sized to CM92's published correlations, 11 of 40 simulated evaluations confirmed, with exact interval 0.146 to 0.439. Pooled no-effect histories gave 3 confirmations in 220. These histories share random streams, so the count is a finite screen, not an independent error rate or a certified size. The reference injected effect gave a mean gain of 0.0045 nats. Barry accepted the finite screen before the first real fit (D022).

S2.7 Access, execution and repair

Stage (8 November 2026, UTC)	Time
Protocol published (DOI 10.5281/zenodo.24053338)	12:34:47
First attempt began; failed before any ledger or score	12:38:28; 12:39:51
Repaired acquisition began	12:51:18
Candidate and reference ledger concordance passed	13:06:58.627
Primary result written	13:06:58.641

The first attempt failed when the archive's literal `uncert="None"` token for missing uncertainty reached a strict parser. Uncertainty enters no extraction, feature, evidence, routing or score path. The repair accepts that exact token without changing or dropping any row. Synthetic invariance tests, independent review, the full test suite, a new full rehearsal and supported-platform CI passed before continuation (D026). All 26 original forecasting source files, the parameter bytes and the protocol bytes are unchanged. The repair freeze is `5e39aeb` and the execution commit `d7151e0`. No person or agent inspected real scores, outcomes, counts or coverage summaries before the primary file existed.

Two independent ledger and evidence implementations had to agree exactly at both delays before any score. A separate checker reproduced all 2,641 variant score rows, with a maximum discrepancy of order 10^{-13}, and the primary interval. Claude recomputed the interval by a separate route. The primary population and routing are in C03.

S2.8 Observation contract, censor repair and redesigned clocks

The original probability calculation is reproducible, but the fitted interval is not itself a known latent-onset set. The R1 contract audit found that clipping may return an envelope when fitted and report intervals conflict, rather than always nesting the former inside the latter. Its crossing kernel also contains a slope-dependent weighting. Normalising that kernel does not prove a generative likelihood for the selected reports. Forecast comparisons using it must therefore be named working scores.

R1 separately audited the first eligible confirmed candidate, including candidates that never issued forecasts. A future rule was needed when a fitted interval straddles a selected stopping cutoff: if issue $< L < C < U$, version v4 uses the working lower-edge backoff while retaining C and its administrative reason. The original sources and scores are preserved. This repair fixes an internal working-set convention; it does not convert statistical bounds into known survival of a physical onset. The actual original ledger has no affected straddle.

The corrected X5 construction searches for decisive faintness after the previous peak before selecting the subsequent decisive-bright rise. It keeps raw time supports and explicit flags. The latent target proposed for R2 is the last upward threshold crossing before decisive brightness. Three premises remain distinct:

- **A0:** correct association, relevant-event completeness and a valid risk set;
- **A1:** containment of the latent crossing in a resolved report window, or genuine survival through a censoring cutoff;
- **A2:** a complete normalised coarsening-at-random observation kernel for the selected data.

Under the relevant premises, the report likelihood can factor into a model-independent observation term and the forecast probability of its compatible onset set. Visit timing alone does not establish that factorisation. Median aggregation can hide a decisive raw report, so exact raw magnitudes and broad time supports do not by themselves establish witness ordering. A margin past threshold is not proof of A0, A1 or A2.

R2b retained the working interval/backoff score and reported a separate deterministic sensitivity. At the realised stopping cutoff C, its mixed functional uses the point-density contrast for a latent onset before C and the survival contrast at C otherwise. Infima and suprema over the compatible report set give conditional sample bounds. They require A0/A1 and genuine survival where used, are not confidence limits, and do not

bound an unknown censored latent-density score. Coverage loss and supersession are selected cutoffs, not independent administrative censoring.

The DF observed clock anchors the current forecast to the first decisive faint report and uses the decisive-bright span as its mark. A delayed issue waits for report availability; a new decisive bright report can abort it. The same scheduled observed-anchor convention defines historical gaps. Prefix extraction reconstructs history at each issue; it is not an immutable catalogue of measurements available under unknown historical upload dates. End interpolation is absent from this clock, but report lag, shared observations and selection remain. An invented cadence construction yields a positive duration-mark gain without physical duration information; it is a counterexample, not an estimate for SS Cyg.

B0 uses previous-gap and older-history covariates, B1 discounted history and B2 a windowed state-space level. Every marked rule adds a development-fitted nonnegative term to its own fixed base and preserves missing-history/mark fallbacks. These models are new windowed comparators, not the original M2. The original-issue, DF-issue and matched point-clock populations are kept distinct. Exact coincident DF clocks share the earliest original representative; later originals remain in the audit. Unsupported outcomes stay unsupported and are not filled with empirical zero gains. The execution freeze is 5fb69c6.

S2.9 Sequential precision experiments and distinct nulls

R2d simulated the learning and availability algorithm in Gaussian point log gaps. Development fits, guarded marks, delayed outcome labels and rolling level updates were rebuilt in each trial. Mark noise, missingness, small development samples, unequal star weights, opposing slopes, shared trends/states and dependence were stressed jointly. Star weights were capped; both star-half and whole-star CR1/t procedures were reported, with whole-star score sign flips as a symmetry-qualified sensitivity. Six equal-count epochs were grouped into halves: these are not actual calendar eighteen-year clusters. No method was selected from the results. The positive-effect cells use a nominal individual latent-oracle gain of 0.004 nats before mark noise, centring and development learning; achieved frozen gains can be much smaller. This is a synthetic input, not C01 or an effect inferred from the archive.

The two statistics ask different questions. In an invented unit-variance Gaussian model with standardised mark x, true coefficient beta and frozen forecast shift b*x,

$$E[U_0] = \beta, \qquad E[\Delta s] = b\beta - b^2/2.$$

Thus a positive coefficient-score moment can accompany negative gain by the actual forecast. With beta=0.05 and b=0.20, expected gain is -0.010 nats. Mark centring, variance centring and nuisance orthogonalisation are different operations; the tested score moment is not asserted to be an efficient or model-free information test.

Strict predictive-null controls used the generator's true conditional mean and innovation variance. No-direct-effect histories with estimated bases or shared states can still contain real incremental predictive information. Their rejection rates are specificity/learning diagnostics, not universally false positives. Known means give realised-path predictable gain and score targets; these are not unconditional long-run KL. Gain-interval coverage was reported with and without exact-zero-offset fits.

The grid fixed 35 cases with 2,000 deterministic trial seeds each. A separate six-star grid fixed eight cases and 16,000 total trials. Exact binomial intervals describe per-cell Monte Carlo uncertainty, not uncertainty about the astronomical process. An observed at-most-five-percent screen does not certify nominal size; neither pooling different null cases nor agreement of same-sample Taylor ratios certifies every full null. Few stars also limit exact sign flips: with four, the minimum one-sided p is 0.0625.

The proposed association-then-gain fixed sequence retains distinct claims. Strong familywise control requires stage one valid under its association null and stage two valid whenever frozen forecast gain is nonpositive, including real association with an oversized trained coefficient. Independence of the stages is unnecessary. Empirical nesting of rejections in the retained grids is not a theorem or a future power guarantee.

S2.10 Gain-free early-record transfer qualification

G3a fixed its protocol and implementation in Git at d827313 before any candidate record was requested. The authorised split is native JD < 2447892.5, with the inclusive provider endpoint strictly below it. Exact names were resolved under the same date restriction; pagination, cap subdivision, identity, row-window and duplicate checks preceded analysis. Every raw response was retained and hashed. Only native Vis./V reports entered qualification; limits remained limits and missing uncertainties remained missing.

The candidate set was SS Aur, RU Peg, AR And, X Leo, TW Vir, UU Aql, CH UMa and DX And, with U Gem and IP Peg as eclipse holds. Thresholds were catalogue proposals, not fitted visual calibrations. A common decisive margin was frozen before access. All candidates needed:

- at least 40 timing-supported, positive-mark DF issuances with an eligible onset window or positive-exposure working censor;
- at least 70% eligible onset windows among confirmed episodes;
- stable nondegenerate unmarked B0 fits at every declared start;
- no operational long-bright-run failure;
- for the holds, an additional frozen eclipse feasibility guard.

Zero-exposure survival terms did not count. B0 convergence used marks stripped and fitted no duration offset. The morphology rule flags a supported threshold-bright run lasting at least sixty days, or repeated forty-day runs, with minimum night/observer support and no long internal observing hole. It is an operational exclusion, not a subtype classifier. The qualifying set was written before any pooled mark fit; it is empty, so no such fit followed.

The holds used two derived observer/JD-night streams: median versus brightest positive estimate, with the strongest numerical bound for all-limit nights. A night is released at its fixed end, not its retrospectively last report. Two-night faint confirmation delays the issue. Both streams must pass ordinary qualification and agree on episode/window/mark and supported-count metrics. A fixed synthetic dip envelope on the same early timestamps is an additional conditional test. Its no-eclipse baseline must first recover enough events with adequate timing accuracy; neither hold passed this baseline, so no eclipse-profile outcome was evaluated. This is not evidence that eclipses caused failure or were removed.

Last-decade supported issuance rates were extrapolated to proposed test horizons solely for planning. G3a required a sufficiently large qualifying cohort and projected yield before a G3b request; neither requirement was met. Candidate test data and SS Cyg 2018 onward remain closed. G3a measured transfer feasibility, not duration effects in other stars.

S3. Results and sensitivities

S3.1 Primary, eras, class rule and sensitivities

Table S1. Frozen comparisons, mean paired working-score gain (nats per issuance) and 90% cluster interval. Only the first row is confirmatory.

Comparison	Issuances	Gain	Interval
Primary, M3 − M2, 1960–2017	441	0.0165	0.0049 to 0.0281
Segment 1960–1992	261	0.0240	0.0091 to 0.0389
Segment 1993–2017	180	0.0057	−0.0107 to 0.0221
12-day class and within-class rule	441	0.0025	−0.0022 to 0.0072
Uniform-end marks	441	0.0166	lower 0.0044
Previous-gap-only baseline	441	0.0170	lower 0.0054
Two-day issue delay	441	0.0165	lower 0.0049
Joint-valid rows	436	0.0167	lower 0.0049
Three-year clusters	441	as primary	lower 0.0059
Baseline M2 − M0	441	0.1168	0.0663 to 0.1672

The primary interval uses ten nonempty blocks and nine degrees of freedom. Eight have positive means. Leave-one-block-out means span 0.0132 to 0.0194, and 0.5397 of individual issuance gains are positive. Holding the estimate, critical value and blocks fixed, the strict failure boundary is approximately 1.42705 times the reported standard error. A 42% inflation passes and an exact 43% fails. None of these checks validates the cluster dependence assumption.

The frozen P3 placebo equals M2, because its development lag coefficient sat at its bound of zero. The planned memory placebo therefore carries no information.

S3.2 Calibration

M3's nominal 50% central intervals covered between 0.546 and 0.617 of outcomes; its nominal 90% intervals between 0.923 and 0.980. Each range runs from definitely covered to possibly covered, given the extracted brackets and censoring, over all scored issuances. These are realised-sample bounds, not confidence intervals for a population rate, and they show overcoverage. Mid-CDF deciles of resolved onsets are descriptive; no uniformity test is claimed.

S3.3 Evaluation-only offsets

Duration offsets were fitted to the evaluation outcomes with an evaluation-refitted M2 held fixed. The support intervals use a likelihood-ratio cutoff of 3.841, with no dependence or measurement calibration.

Offset	Estimate	Conditional support
Current duration, separate	0.04209	0.01574 to 0.06845
Previous duration, separate	−0.03836	−0.06489 to −0.01184
Current duration, joint	0.03045	−0.00063 to 0.06152
Previous duration, joint	−0.02213	−0.05340 to 0.00913

The joint design is correlated but has full rank, with condition number 1.79. Both joint support intervals include zero. Correlation alone neither explains that nor makes the coefficients nonidentifiable.

S3.4 A slow component can hide behind negative lag signs

Negative lag signs alone cannot exclude a dominant slow driver (A10k). Take independent unit-variance AR(1) components S and A, with persistence 0.95 and −0.9, and set log duration $= 0.175S + 0.350A$ and log gap $= 0.300S + 0.001A$. The lag-one duration correlation is then about −0.53, and the lag coefficient conditional on the preceding gap is negative. Yet S dominates gap variation and the current conditional covariance. This construction is a logical counterexample, not fitted to SS Cyg and not a rival empirical result.

S3.5 Exploratory post-result calculations

Everything in this subsection reuses the evaluation outcomes. Each interval conditions on the fitted adjustment it describes. None supplies a further held-out confirmation.

- **Level shift and recentring (C20).** The best common M2 log-location shift is −0.04679. It reduces the original untruncated clock scale by about 4.57%. The frozen duration term implies a level shift of −0.01108, because the evaluation mean standardised log duration is −0.2379. Shifting M2 by that implied amount accounts for 0.00370 nats (0.00013 to 0.00726). The contrast after recentring the duration at its evaluation mean is 0.01283 (0.00359 to 0.02206). With both models at the best M2 level, the gain is 0.01239 (0.00098 to 0.02381). The two orderings of the split differ by about 0.000012 nats.
- **Class versus continuous duration (C21).** With M2's historical parameters fixed and a free common level, the evaluation likelihood-ratio comparisons are as follows:
  - class versus level only, 14.7;
  - continuous log duration versus level only, 11.7;
  - within-class duration added to class, 0.17.

  The class contrast corresponds to a 14% larger untruncated working-clock scale after long outbursts. In 1910–1959 the class and continuous fits were similar, with a smaller contrast.
- **Shared level for the class rule (C22).** The frozen class-plus-within rule's mean offset is 0.01728. Giving both rules M2's evaluation-fitted level yields a gain of 0.00905 (0.00409 to 0.01401). For 1993–2017 alone it yields 0.00761 (−0.00631 to 0.02153). The uncentred type component cannot shorten short-outburst forecasts, but the full frozen rule does shorten 181 of 245 of them through its within-class term.
- **Era refits (C23).** The class contrast is 0.16323 (statistic 13.504) in 1960–1992 and 0.07828 (2.262) in 1993–2017. These are descriptive; there is no test of change.
- **Alternation (C24).** Across forecast-record pairs, the lag-one correlation of log duration is −0.5233 in the evaluation years and −0.4006 in the development years. A long outburst followed a long one in 0.1495 of pairs and a short one in 0.6640. The pairs include one repeat-flagged record and are not a catalogue of unique consecutive outbursts.

S3.6 Development fit

In 1910–1959 the duration offset was 0.0466 per standard deviation, with likelihood-ratio statistic 5.1935 against M2. This fit is in-sample and overlaps CM92's record, so it is not evidence for the claim.

S3.7 Prior art checked against the source

Claude re-read CM92's Table 4 and its Fig. 5 subscript conventions on the original scan (Cannizzo and Mattei 1992) . For 705 outbursts (1896–1992), the correlations are:

- duration with the following quiescence, 0.223;
- duration with the preceding quiescence, 0.282;
- successive durations, −0.400.

Durations split at 12 days, with 401 shorter and 304 longer. The correlation of duration with the following onset-to-onset cycle is 0.475, against 0.145 with the preceding cycle. For end-to-end cycles the pattern reverses, at 0.093 and 0.521. The famous relation is therefore largely a consequence of the cycle definition, which already contains the duration; CM92 showed that cycle definitions matter.

S3.8 Bounded independent novelty check

An independent search in A10m found no prior preregistered probabilistic held-out recurrence evaluation, or explicit held-out comparison of duration beyond a timing-history baseline, among the studies located and checked. This supports a bounded "to our knowledge" statement, not proof of universal priority. The exact queries and source receipts are in `data/provenance/a10m_novelty/`; the appraisal is `docs/audits/2026-11-09-a10m-novelty-review.md`.

The closest precedent, Grigorievskiy et al. (2014), uses chronological training, validation and test sets, with one test outburst of V363 Lyr. It scores point-onset errors; no preregistration or scored predictive onset distribution is reported. No explicit duration-addition comparison against a common timing baseline is reported. Repeated random fits do not make the evaluated forecast a probability distribution. The independent search included an OpenAlex citing-title screen and primary-source checks of this study and Price et al. (2007). ADS access failed; observer/non-English coverage is incomplete, and some full texts, including Tsuranova et al. (2026), remain inaccessible. Their unverified methods are not treated as exclusions.

S3.9 R1 rescoring and endpoint-error diagnostics

The censor audit's future repair changed zero original outcomes. Deleting all six censored issuances gives 0.01678 nats on the remainder, a descriptive deletion sensitivity rather than a bound on every observation-model correction. The date-selected Figure 1 example has model median remaining waits 27.5749 and 26.7726 days and probabilities within thirty days 0.576727 and 0.602765. It does not estimate mean observational utility.

Saved-point density scoring gained 0.015404 nats (0.003443 to 0.027365) on 441 issuances. Its points are fitted summaries. Original report clips were missing for 434 associations. Only 11 contrasts, including five exact fallback zeros, supported the proposed report-interval/kernel sensitivity. That selected subset is not a full-population replication. The later X5 repair supplied 818 available and 805 eligible windows among 845 confirmed episodes, with median width 1.1495 days; it did not retroactively alter the frozen result.

The endpoint-dose study used 80 histories under null and gap-dependent scenarios. Independent Gaussian errors were added to fitted ends after extraction, while issue and onset times stayed fixed. Every tested all-issued scenario/dose mean was nonpositive. Large doses disabled landmarks and increased exact common fallbacks, so zero gain did not certify latent measurement validity. Truth-anchor comparisons selected completed, still-quiescent matched emissions. Systematic/state-dependent errors, regenerated observing behaviour and unseen waveforms were not covered. One floating-point concordance failure and its separately frozen same-seed recovery remain in the execution record; no scientific predictor or gain was tuned from that failure.

S3.10 R2b report-window and clock contrasts

Table S2. Exploratory R2b headline contrasts. Every six- and twelve-year interval crosses zero. These intervals describe the stated working-score populations; they do not validate physical onset containment, coarsening at random or prospective coverage.

Exploratory contrast	Scored issuances	Mean gain, nats	Six-year 90% interval	Twelve-year 90% interval
H1: fitted-end clock, B2 duration	432	0.0075	−0.0008 to 0.0159	−0.0014 to 0.0165
H2: observed clock, B2 bright span	431	0.0043	−0.0008 to 0.0094	−0.0021 to 0.0107
H3: observed-clock gain minus matched fitted-end gain	431	−0.0031	−0.0078 to 0.0017	−0.0078 to 0.0017

Original evaluation issues numbered 441; the unique delayed-clock schedule numbered 440. Those are not the scored denominators. Missing windows, candidate obstacles and invalid risk sets stayed explicit; they were not skipped to credit a later event. H3 is a paired difference between incremental working gains, not a causal measurement-error effect.

The fixed-C mixed sample envelope for H2 runs from 0.0009703 to 0.0077535. Its positive lower mean is conditional on the report/risk-set and survival premises. It is not a positive confidence limit or proof of latent forecast improvement. The observing-rate proxy gained 0.0030 under the working rule, but its duration-dependent span prevents treating it as a validated cadence negative control. A one-lag anchor-shift Fisher index is only local: independent exact integration exceeds its quadratic approximation in an invented example. It cannot serve as an upper bound ruling out cadence. Midpoint residual and block-dependence proxies are likewise selected, noisy diagnostics rather than validated future-star covariance.

All actual model scores were reconstructed by separate numerical routes. That agreement establishes implementation concordance, not parameter identifiability, measurement calibration or the observation premises.

S3.11 Baseline adequacy and post-result rolling guards

On the DF development population, negative log likelihood was 830.19 for B0, 827.24 for B1 and 849.28 for B2. B2 was preselected, not empirically strongest. Its lower fit motivates review of its particular windowed state implementation; it does not uniquely establish the physical recurrence process. Measured geometric mean gaps in the first evaluation blocks were shorter than the development-block range, without resolving physical versus observational causes.

Historical guards subtracted the preceding ten marks and used rolling recalibration from preceding available evaluation outcomes. They retained the unguarded development coefficient, used evaluation-only history and final saved labels at proxy resolution times. Simple closure and coverage-recognition delays left update sets unchanged in SS Cyg, but complete confirmation, recrossing and eligibility availability was not reconstructed.

The point-original B0 duration gain changed from 0.0184 to 0.0132. The rate gain changed from 0.00304 to 0.00142; the guarded estimate remained positive and inconclusive. Guarded DF B0/B2 duration intervals also crossed zero, while point-B0 retained positive lower limits under both blockings. Changing base and mark together creates an ordering-dependent descriptive sensitivity, not a causal allocation or a measured fraction of C01. Different significance labels do not establish a predictor difference.

Sequential trend counterexamples show that rolling guards can leave positive score moments and even genuine gain over a lagging base without a direct physical mark effect. The same-sample score/gain Taylor ratios reproduce, but shared scores and curvature enter both sides. They are not independent prospective power evidence.

S3.12 Multi-star precision and sequence diagnostics

Table S3. Whole-star rejection frequencies at a nominal individual latent-oracle gain of 0.004 nats, before noise, centring and learning. The issuance counts are nominal; stressed cases lose usable mark support. Percentages concern each complete generator/learning process, not a common supported sample size or achieved gain.

Stars and condition	Nominal evaluation issuances	Score-moment rejection	Frozen-gain rejection
Eight, clean	1,100	81.8%	36.7%
Eight, persistent / small development / missing marks	1,100	63.6%	—
Eight, noisy / delayed / common state	1,100	38.7%	—
Six, clean	1,100	76.8%	33.1%
Six, clean	2,000	94.6%	48.8%
Six, persistent / small development / missing marks	1,100	59.4%	21.3%
Six, noisy / delayed / common state	1,100	36.2%	10.3%

All 70,000 original-grid and 16,000 supplemental trials completed without failed fits. Neither grid establishes a robust sample size across its stressed envelope. The six-star supplement has estimated-base no-direct-effect cells, not separate six-star strict-oracle controls.

Gain never rejected without the score test in either retained grid. Independent Gaussian arithmetic disproves general nesting, so this is not a prospective power identity. The pooled original strict-oracle score rate was 397/8,000; pooling does not certify every scenario or the partial null of real association with nonpositive gain. An exactly five-percent procedure can fail an observed-at-most-five-percent Monte Carlo screen frequently; the frozen screen is retained as the finite rule it was, with uncertainty, not reinterpreted as proof of miscalibration or replaced by a pooled pass.

S3.13 Transfer qualification, attrition and the SS Aur flag

Table S4. Frozen G3a qualification. Hold-star counts use the median derived-night stream with delayed confirmations, not the ordinary raw-report diagnostic.

Candidate	Supported development issuances	Eligible onset windows	Main failed conditions
SS Aur	110	79.6%	Operational morphology
RU Peg	23	63.5%	Count, windows
AR And	7	31.4%	Count, windows, base information/fit
X Leo	35	44.0%	Count, windows
TW Vir	0	11.6%	Count, windows, base information/fit
UU Aql	15	80.8%	Count
CH UMa	0	70.4%	Count, base information/fit
DX And	0	70.8%	Count, base information/fit
U Gem, hold	27	90.6%	Count, eclipse guard inconclusive
IP Peg, hold	0	66.7%	Count, windows, base information/fit, eclipse guard inconclusive

Across frozen streams there were 217 supported development issuances. The frozen 1980s streams supplied 102 supported issuances; all candidates ignoring gates project 367.2 over 36 years if that rate persists. No actual test count was opened.

The independent ordinary raw-report diagnostic must remain separate. It gave 105 supported 1980s issuances against 485 confirmed episode peaks, compared with SS Cyg's 67/73. The ratios count issues and peaks by different time landmarks and are not paired recovery probabilities. In this raw diagnostic, 822 clock-available development issuances fell to 280 with a covered previous gap and 274 after requiring a mark. Outcome support reduced the corresponding covered-gap unmarked/marked totals to 225/221. Missing covered history was the largest counted loss; the causes include observing holes, window eligibility and candidate association, not a unique depth effect.

Deleting history and mark gates in the raw diagnostic gave 201 1980s opportunities and at most five stars with forty development opportunities. At that assumed rate the 36-year projection is 723.6. This does not meet the proposal, and it is not the frozen hold-stream result or an impossibility bound for other models. We omit speculative later split projections because they do not measure later observing coverage.

Whole-pre-1990 median fade-confirmation delays were longer in candidates than the SS Cyg reference. These measure the closure of the last threshold-bright visit to the last raw support time of the first selected decisive-faint visit, not true-end timing errors. They do not justify allocating the forecast loss to photometric precision.

The SS Aur flag spans 100.97 days. Separate native parsing and graph reconstruction found 258 votes, including 252 positive visit medians; their magnitude quantiles were 12.5, 13.2 and 13.7. The exact nondecisive threshold-bright band, $12.5 < m \leq 13.5$, contains 176/252. Its decisive-bright nights form four separated groups, with one, seven, three and five nights. This is not an independently annotated physical-outburst count. The record-wide 14.5-mag median is selected from reports fainter than threshold and pools other years and observers. No sequence change, observer offset, intrinsic brightening, contamination or subtype is adjudicated. Removing this morphology failure would still not create the required cohort under the unchanged count gates.

The expansion stops for this paper. Preserving the empty qualifying set and untouched test records records what the chosen transfer failed to support. It does not establish absent duration information in other stars or the impossibility of a materially different analysis.

S4. Analyses superseded during review

Each entry records what was replaced, who found the problem, and where the record is.

Earlier analysis or statement	Problem and finder	Replacement	Record
Duration predicts the following onset-to-onset cycle	Shared term with the cycle (CM92); confirmed in the A01 audit	Quiet-gap target only	A01 memo
Extractors X1, X2, X3	Failed synthetic qualification (Codex and Claude reviews)	X4, qualified at magnitude 10 only	A04e–A04i
Unrestricted duration coefficient on all routes	End-time coupling and route mixing (Claude, D020)	$\beta \geq 0$, matched routing, local gap level	D020, D021
Joint refit of M3 with M2	Training loss not matched to deployment (Codex, D021)	Mark offsets fitted with M2 fixed	D021
Candidate-authority concordance rule	Allowed mismatch deletion (Codex, D024)	Exact two-implementation concordance or abort	D024
Strict parser for the uncertainty field	Failed on `None` after access (execution)	Exact-token repair, reviewed and rehearsed	D026

Interpretation corrections checked in A10k. Codex's independent audit narrowed several readings from Claude's A10j review before the products were drafted. Claude accepted all of them (D030).

Earlier reading	Corrected scope	Evidence
Global recentering isolates drift-free information	It defines an approximately additive descriptive score contrast; within-period drift remains possible	
All predictive information is in outburst type	A class indicator has greater in-sample likelihood; within-class equivalence was not tested	
The entire frozen class rule cannot shorten short-event forecasts	That restriction concerns its type component; the within-class component can shorten them	
Negative lag excludes a dominant slow driver	Alternation and a slow component can coexist; no mechanism allocation follows	
A rounded SE boundary is a safe inclusive margin	The decision is strict; the rescaling boundary is exclusive and does not validate dependence	
Frozen M2 forecasts are 4.6% too long	The best level shift reduces the untruncated clock scale by 4.57%; that is not a remaining-wait bias	
The gap after a long outburst is about five days longer	A 14% clock-scale ratio; days need an explicitly chosen reference scale	

The original exploratory numbers are retained. These corrections do not alter the frozen primary comparison, its outcomes or the predeclared class-rule result.

Novelty wording narrowed in A10l. The draft framing called this the first preregistered held-out forecast test of dwarf-nova recurrence. The refreshed search found a held-out point-forecast study of V363 Lyr by Grigorievskiy and colleagues, in the 2014 ITISE proceedings (pp. 520–531), with a single test outburst. The claim now specifies preregistered, probabilistic and duration-based.

Draft corrections in A10m. Semantic review corrected an overstatement of complete flare recovery, distinguished an unseen-waveform qualification failure from unperformed X4 recovery, defined the standardised timing predictors, and gave the within-class likelihood statistic its proper class-model reference. The article's assumptions table now states calculated results and current limits rather than attributing information exclusively to class or physical disc state. The uniform-end lower bound is rounded directly from the full-precision evidence, and the SE failure boundary is explicitly approximate when printed. Reproduction commands preserve committed evidence and state their retained-input dependencies. The primary comparison is unchanged.

Observation, baseline and transfer corrections after external review.

Earlier reading or implementation	Reviewed replacement	Evidence
X4's fitted interval is a hard observing bracket or calibrated posterior	Working crossing summary; proper latent-onset observation likelihood remains unestablished	R1 contract audit
The censor rule handles every interval straddle	Separately versioned lower-edge backoff; zero original outcomes changed	
Two-sided report rescoring is available for the original full population	Missing lower clips prevent that; corrected X5 is a new construction	
An observed clock eliminates endpoint/cadence mechanisms	It removes one literal fitted end but retains report lags, selection and shared observations	
B2 is the strongest activity comparator	It was preselected; development and descriptive held-out fit were weaker	

Earlier reading or implementation	Reviewed replacement	Evidence
Guards identify an era component of C01 or remove all slow covariation	They change both rules and leave other trends; no causal share of C01 was measured	
Positive score detection confirms trained forecast gain	Distinct full nulls; overshoot counterexamples and no universal sequence nesting	
Candidate/reference ratios are paired cycle-conversion probabilities	Issuance/peak descriptive rates; ordinary and derived hold streams are separate	
SS Aur's flag establishes four outbursts or photometric contamination	Four report-night clusters and threshold-proximate medians, with no cause adjudicated	
Frozen transfer failure excludes every later model or data source	Failure of the specified screen; stopping expansion is a scope judgment	

None of these reviews retunes or overwrites the original comparison. The failed protocols, source freezes, bounded access receipts and negative/inconclusive results remain in the record.

S5. Reproduction and navigation

From the repository root, without new archive requests:

These public-only checks use the committed aggregates and figure assets:

```
python3 manuscript/figures.py --check
make -C manuscript check
make -C manuscript
```

Recomputing scores and exploratory fits requires the retained inputs outside Git. The saved-score checker needs the completed evaluation run. The three subsequent analyses also need the retained 1910–1959 development inputs. Outputs below go to scratch paths; committed evidence is preserved.

```
python3 scripts/check_a10g_evaluation.py \
  --directory outputs/a10h_live_repair_5e39aeb \
  --output outputs/a10m_reproduction/score_audit.json
uv run --frozen --extra analysis \
  python scripts/a10j_postresult_review.py \
  --output outputs/a10m_reproduction/a10j_summary.json
uv run --frozen --extra analysis \
  python scripts/a10k_exploratory_audit.py
uv run --frozen --extra analysis \
  python scripts/a10k_secondary_audit.py
```

The retained run includes raw archive replies, observer identities and per-issuance scores. Its archive SHA-256 and manifest are in `data/provenance/a10i_evaluation_1960_2017/archive.json`. The committed aggregates support the ledger values, but do not replace these inputs for a full recomputation. The published protocol is at https://doi.org/10.5281/zenodo.24053338.

The observation/transfer reviews use the same retained early records and saved outputs. These commands write new audit outputs and do not download data:

```
uv run --frozen --extra analysis python scripts/r1_observation_scores.py \
  --freeze-commit 0594146 --output outputs/paper_reproduction/r1_scores
uv run --frozen --extra analysis python scripts/check_r2b_execution.py \
  --run outputs/r2b_real_5fb69c6 --allow-real --output outputs/paper_reproduction/r2b_check
uv run --frozen --extra analysis python scripts/check_r2d_review.py \
  --output outputs/paper_reproduction/r2d_review
uv run --frozen --extra analysis python scripts/check_g3a_execution.py \
  --output outputs/paper_reproduction/g3a_check
uv run --frozen --extra analysis python scripts/check_ss_aur_flag.py \
  --output outputs/paper_reproduction/ss_aur_flag
```

The endpoint-dose and precision-grid regeneration commands, individual deterministic seeds, source locks and recovery receipts are in the R1, R2d and G3a audit/protocol files. Full regeneration is distinct from recounting saved trials and is unnecessary for rebuilding the paper. Public evidence resides in data/provenance/r1_20261109, r2b_20261109, r2d_precision, g3a_preparation, g3a_20261110 and g3a_review_independent. These aggregates do not replace native inputs for an independent full measurement reconstruction. No command in this guide authorises a new candidate test-period request or SS Cyg 2018+ access.

Navigation:

- `manuscript/claims.csv`: every claim, with its status, evidence, commit, reproduction command and limits.
- `manuscript/estimands.md`: allowed and forbidden readings.
- `coordination/decisions.md`: D013–D047.
- `docs/audits/`: every review, including A10i (the result), A10j (Claude's review), A10k (Codex's audit) and A10l (sources and drafts).
- `docs/outcome-access-ledger.md`: who saw which outcomes, and when.
- `manuscript/text/for-agents.md`: the agent guide.

References

Cannizzo, John K., and Janet A. Mattei. 1992. "On the Long-Term Behavior of SS Cygni." *The Astrophysical Journal* 401: 642. https://doi.org/10.1086/172092.

Price, A., A. A. Henden, G. Foster, V. Petriew, R. Huziak, R. James, M. D. Koppelman, et al. 2007. "SS Cygni Outburst Predictors and Long Term Quasi-Periodic Behavior." *Publications of the Astronomical Society of the Pacific* 119 (862): 1361–66. https://doi.org/10.1086/524413.

Supplementary Information

Fig. S1.

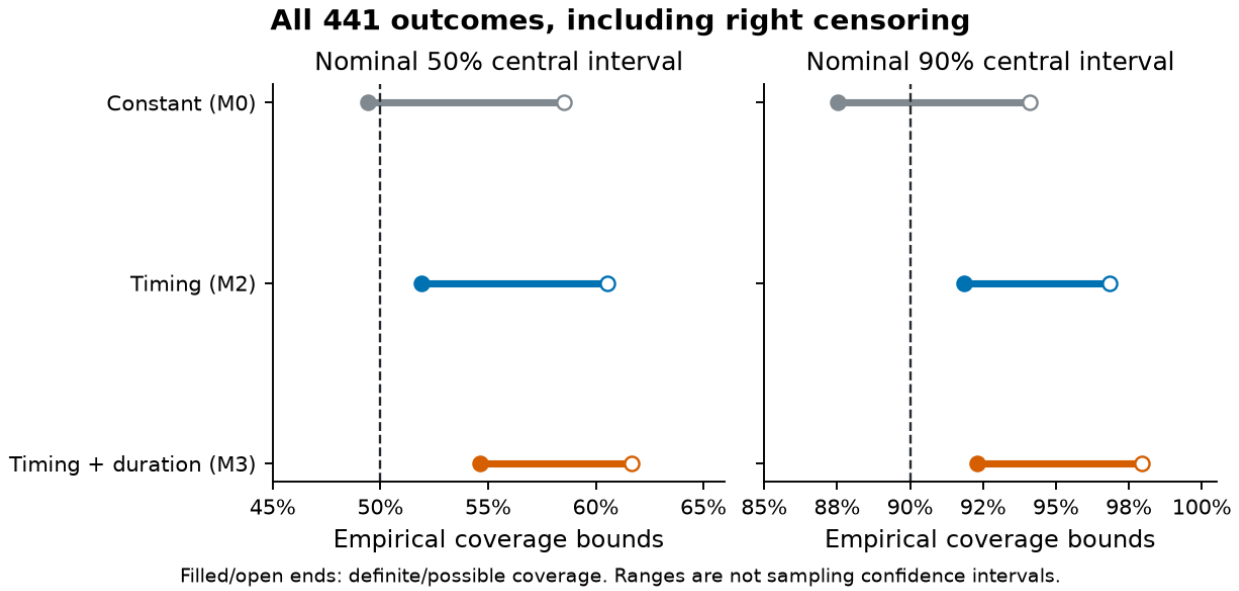


Figure S1: Observed coverage of central predictive intervals for the constant, timing and timing-plus-duration forecasts. Each segment runs from definite to possible coverage given the extracted onset bracket or right censoring; filled and open endpoints distinguish these bounds. Dashed lines mark the nominal fractions. These are partial-identification bounds in the realised sample, conditional on the extraction and scoring population. They are not confidence intervals for population coverage or guarantees about physical onset times. The sample shows overcoverage; the primary paired gain does not imply sharply calibrated operational predictions. Generated by `manuscript/figures.py`.

Fig. S2.

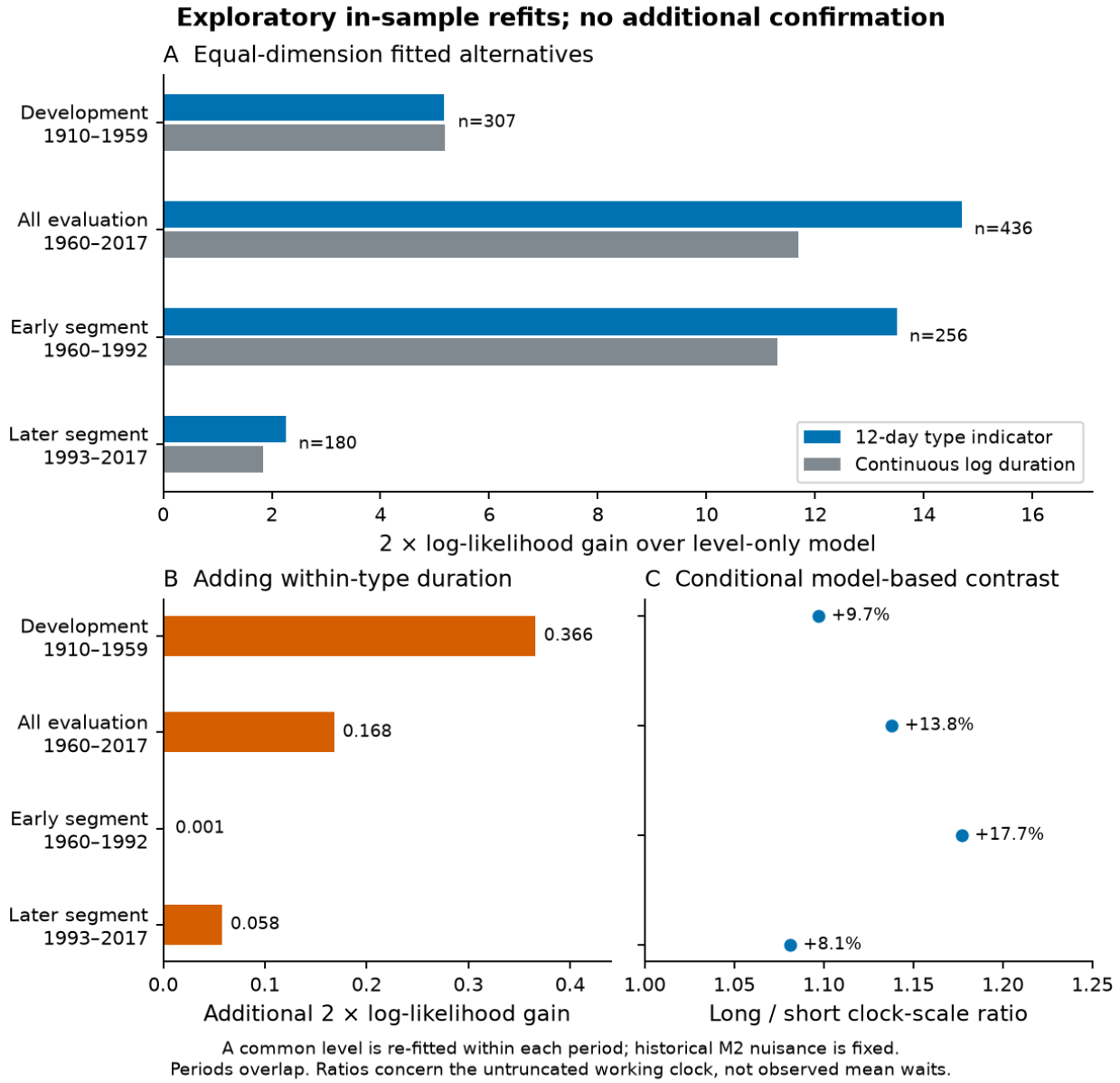


Figure S2: Exploratory post-result comparisons, fitted using the evaluation outcomes. With a free level and historical timing nuisance parameters held fixed, the class-only fit has likelihood-ratio statistic 14.7, compared with 11.7 for continuous log duration; these equal-dimensional models are nonnested. Adding a linear within-class term to class gives 0.17. This is neither exclusive attribution to class nor an absence or equivalence test for within-class effects. The right panel gives conditional long/short ratios of the untruncated working-clock scale, rather than observed-gap or remaining-wait ratios. Era points are descriptive refits, not an era-difference test. No panel supplies a further held-out confirmation. Generated by `manuscript/figures.py`.